EE3A Rivo Computer Organization HW1

1. Yield=(1+(defect*die_area/2))^-2

   wafer_A cost=12, dies=84, area=15*15*pi/4=225pi/4cm^2, defect=0.02/cm^2

   wafer_B cost=15, dies=100, area=20*204/=100picm^2, defect=0.035/cm^2

   (a) Yield_A=(1+(0.02*225pi/8/84))^-2=0.9592

   Yield_B=(1+(0.035*100pi/100/2))^-2=0.8985

   (b) dies*1.1, defect*1.1

   die area=100pi/110=2.856cm^2

   Yield_B=(1+(0.035*1.1*100pi/110/2))^-2=0.8985

2. total power=Pdyn+Pst

   Dynamic_power=capacitive_loads*V^2*f

   Static_power=leakage_current*V

   Processor_A f=3.2GHz, V=1.2V, P=10W+80W

   Processor_B f=4GHz, V=0.8V, P=20W+50W

   (a) CL_A=80/1.44/3.2G=17.36*10^-9C/V

   CL_B=50/0.64/4G=19.53*10^-9C/V

   (b) A: static_power=10/90=11.11%, static/dynamic=1/8=0.125

   B: static_power=20/70=28.57%, static/dynamic=2/5=0.4

   (c) A: 81=80x^2+10x, x=0.9457, V=1.1348 reduced 5.43%

   B: 63=50x^2+20x, x=0.9402, V=0.7522 reduced 5.98%

3. Clock cycle=CPI*IC

   CPU_time=CPI*IC*Clk_Period

   P1 2GHz CPI=1,2,3,4,5

   P2 2.5GHz CPI=2,3,3,4,4

   Program: IC=10^8, (0.2,0.25,0.3,0.1,0.15)

   (a) Time_1=(0.2+0.5+0.9+0.4+0.75)*10^8/(2*10^9)=0.1375

   Time_2=(0.4+0.75+0.9+0.4+0.6)*10^8/(2.5*10^9)=0.122→faster

   (b) P1 CPI=2.75

   P2 CPI=3.05

   (c) P1 clock_cycle=2.75*10^8

   P2 clock_cycle=3.05*10^8

4. CPI: arithmetic=1, load/store=12, branch=5

   IC: arithmetic=2.56*10^9, load/store=1.28*10^9, branch=256*10^6

   (a) reduce amount of load/store CPI for *0.5 time

   time=(2.56+12*1.28+5*0.256)*10^9/f=19.2G/f

   for time=9.6G/f=(2.56+?*1.28+5.0.256), ?=4.5

   ⇨ load/store CPI=4.5, (12-4.5)/12*100%=62.5% => reduce load/store CPI by 62.5%

   (b) time_improved=(0.5*2.56+0.8*12*1.28+0.5*5*0.256)G/f=14.208G/f

   time reduced by (19.2-14.208)/19.2*100%=26%